

Pratham Bhanushali

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PROFESSIONAL SUMMARY

Cybersecurity professional and M.Tech candidate focused on offensive security, vulnerability assessment, and security research. Hands-on experience with web and mobile application penetration testing, network reconnaissance, vulnerability analysis, and Python-based security automation. Experienced with Burp Suite, OWASP ZAP, Metasploit, Nmap, Wireshark, reverse engineering tools, and malware analysis platforms. Strong interest in adversary emulation, MITRE ATT&CK, attack-path analysis, and developing security tooling to identify and validate enterprise security weaknesses.

EDUCATION

National Forensic Sciences University

M.Tech in Artificial Intelligence and Data Science (Cybersecurity)

Gandhinagar, Gujarat

June 2024 - July 2026

- **Relevant Coursework:** Network Security, Malware Analysis, Digital Forensics, Cryptography, Information Security Management Systems, Risk Assessment & Mitigation

Silver Oak University

B.Tech in Computer Engineering

Ahmedabad, Gujarat

July 2020 - June 2024

- **Relevant Coursework:** Networking, Operating Systems, Artificial Intelligence, Machine Learning, Databases

TECHNICAL SKILLS

Offensive Security: Vulnerability Assessment, Penetration Testing, Web Security, Network Reconnaissance, OWASP Top 10, MITRE ATT&CK, Security Testing

Security Tools: Burp Suite, OWASP ZAP, Metasploit, Nmap, Wireshark, Nessus, MobSF, Frida, JADX, APKTool, SQLMap

Programming & Scripting: Python, Bash, C++, SQL, Dart; Security Automation and Scripting

Malware & Analysis: VirusTotal, Any.Run, Hybrid Analysis, Reverse Engineering, Static/Dynamic Analysis

Network & Security Monitoring: TCP/IP, Network Monitoring, Wireshark, Splunk, Vulnerability Scanning, Network Segmentation

AI & Security Research: Adversarial ML, Model Security Auditing, TensorFlow, PyTorch, Scikit-learn, LangChain

Platforms & Tools: Docker, Git, FastAPI, MLflow, Labshock, Claroty/Nozomi concepts

Professional Skills: Analytical Thinking, Problem Solving, Technical Documentation, Team Collaboration, Security Research

PROJECTS

- **AI based Kill-Switch for OT Security** | *Labshock, Docker, DNP3, EtherNet/IP, S7*
Developed a real-time defensive mechanism for ICS environments using hardware-in-the-loop simulation. Analyzed DNP3, EtherNet/IP, and S7 traffic to identify anomalous communication patterns. Implemented automated model switching, drift detection, and response mechanisms for security testing.
- **Email Analysis AI Agent** | *Crew AI, VirusTotal, Hybrid Analysis*
Developed a multi-agent security workflow for automated phishing and malicious-email analysis. Integrated behavioral analysis and threat intelligence lookups to assess indicators and determine severity. Automated security alerts and analysis through agent-based orchestration.
- **MLOps for Secure Model Deployment** | *Python, FastAPI, Docker, MLflow, Grafana*
Designed a secure MLOps pipeline with CI/CD integration, model tracking, containerization, monitoring, and automated security scanning. Applied security controls across model deployment and supporting infrastructure.
- **AI-Powered Intrusion Detection System (IDS)** | *Python, TensorFlow, Scikit-learn, Wireshark*
Developed a machine-learning-based IDS to detect anomalous network traffic and suspicious behavior. Analyzed network and log data using deep learning models to identify potential security breaches and abnormal activity.
- **AI in VAPT (Smart Vulnerability Scanner)** | *Python, Scikit-learn, OWASP ZAP, Burp Suite*
Designed an AI-assisted vulnerability scanner for automated web application security testing. Implemented vulnerability classification, risk prioritization, and automated security reporting to support penetration-testing workflows.
- **Android Application Pentesting Toolkit** | *Frida, MobSF, Drozer, JADX, APKTool*
Performed security testing of vulnerable Android applications, identifying insecure data storage, hardcoded secrets, and exposed components. Used static analysis, dynamic instrumentation, and reverse engineering tools to investigate application behavior.
- **Web Application Pentesting** | *Burp Suite, OWASP ZAP, SQLMap, Metasploit*
Performed hands-on penetration testing of vulnerable web applications, identifying and validating SQL injection, XSS, CSRF, and file-upload vulnerabilities. Documented attack evidence, impact, and remediation recommendations based on OWASP Top 10.
- **Document Chatbot** | *Python, LangChain, Transformers, OpenAI API*
Built an NLP-driven chatbot capable of answering domain-specific queries from uploaded documents. Integrated LLMs and LangChain for semantic retrieval, contextual understanding, and automated responses.

COURSES AND CERTIFICATES

- Google Cybersecurity Professional Certificate (Coursera)
- Let's Defend: SOC Analyst Learning Path
- QRadar SIEM: Understanding the user interface and architecture (v7.5) (IBM)